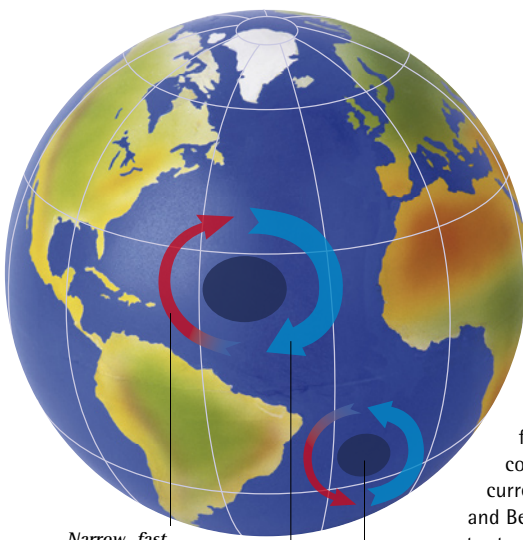




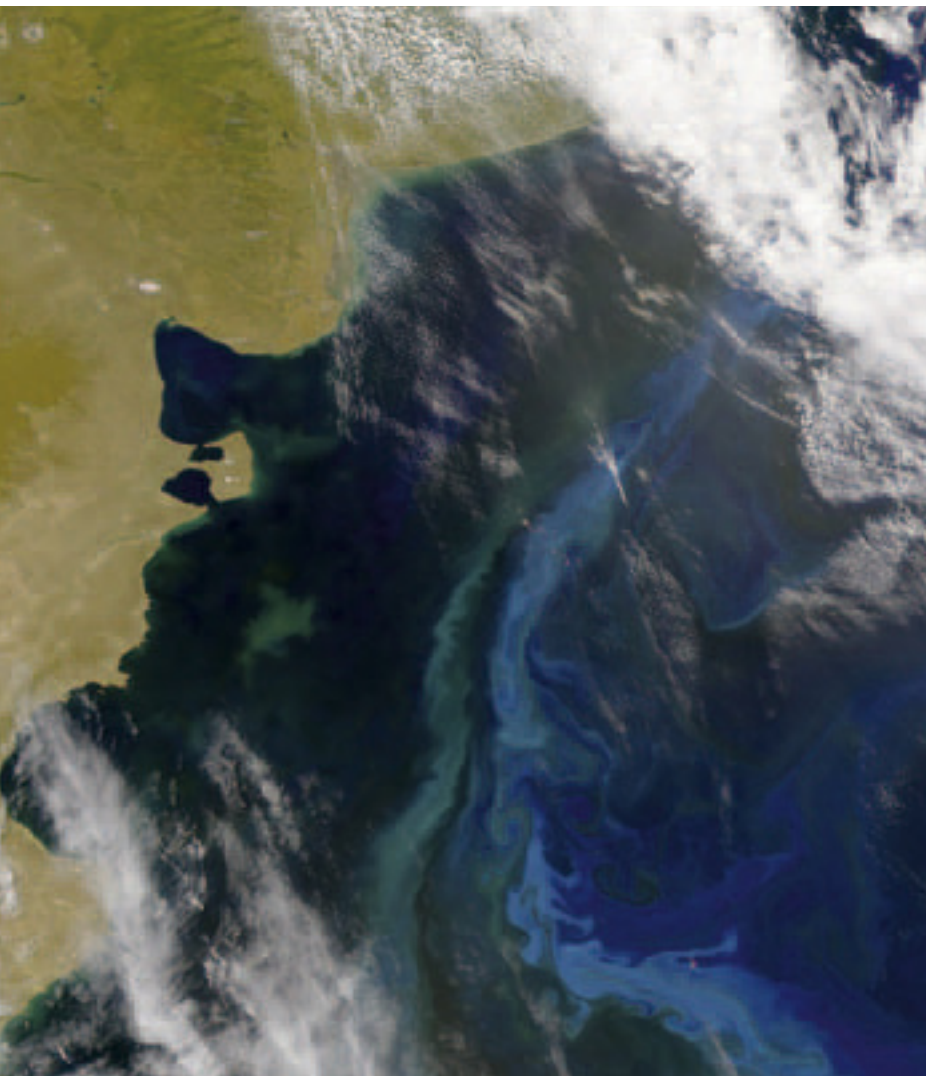
Narrow, fast,
warm western
boundary current

Broad, slow,
cool eastern
boundary current

Still zone at
center of gyre

◀ BOUNDARY CURRENTS

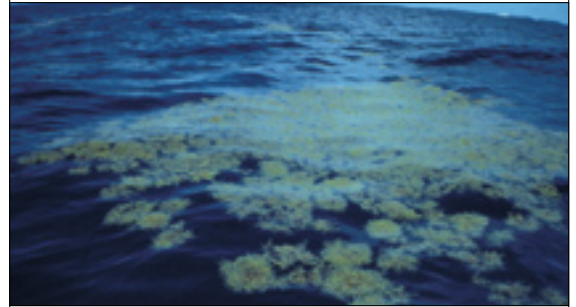
Equatorial currents flow westward in the Atlantic and Pacific, so ocean water tends to pile up against the continents. This forces warm water north and south away from the tropics in powerful western boundary currents such as the Gulf Stream and the Kuroshio Current. These flow faster than the broader, less concentrated eastern boundary currents, such as the California Current and Benguela Current, which carry cold water toward the equator.



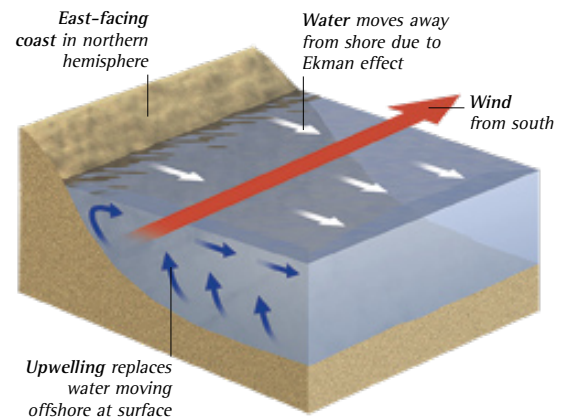
▲ WHERE CURRENTS MEET

Where warm and cold currents meet, the cold water tends to push beneath the warm water and stir up minerals from the ocean floor. These contain vital nutrients that fuel the growth of marine life such as plankton. Often the water in the two currents does not mix easily, as seen in this satellite view of the cold Falklands Current on the left, carrying green plankton, encountering the warm, southward-flowing Brazil Current carrying blue plankton.

STILL CENTERS

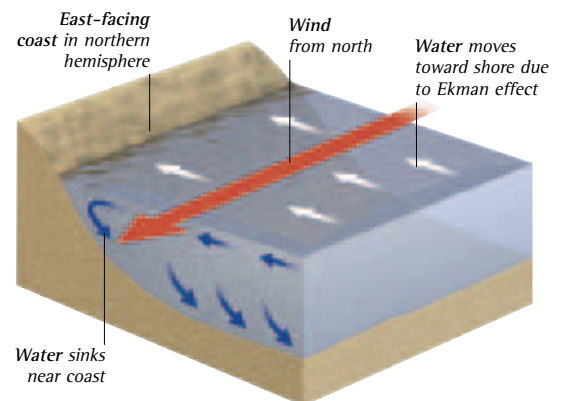


The Ekman spiral pushes ocean water toward the still zone at the center of a current gyre, heaping it up in a broad, shallow mound. In the Sargasso Sea at the center of the North Atlantic gyre, the water level in the middle of the calm zone is about 3 ft (1 m) higher than the water level at the edge. The ocean surface is also covered with floating seaweed called sargassum weed, which has been concentrated in the Sargasso Sea by converging currents.



▲ UPWELLING ZONES

Where the wind and Ekman spiral drive surface water away from the shore, deeper water rises to take its place. This draws nutrients up from the ocean floor, causing blooms of plankton that feed marine life. These regions are called upwelling zones.



▲ COASTAL DOWNWELLING

The opposite of upwelling happens on coasts where the wind and Ekman spiral drive surface water toward the shore. This forces it to sink, suppressing the flow of nutrients toward the surface. As a result, there is little marine life in such downwelling zones.